

▪ Residência unifamiliar

▪ Cargas

⇒ Permanentes:

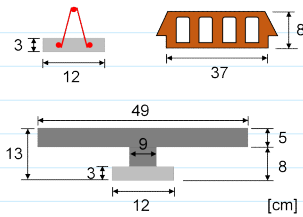
l, Peso próprio: 2,07 kN/m²l, Revestimento: 0,50 kN/m²⇒ Variável: 1,5 kN/m²

⇒ LT 13 (8+5)

⇒ Tabela H8 - 8 x 37 x 20

⇒ Concreto C25

l, Agregado basalto



$$I_{xc} = \frac{b h^3}{12}$$

$$I_{x'} = \frac{b h^3}{3}$$

Vãos do colado: 3,85 m, simplesmente apoiada

ELS do Deslocamento Excessivo

$$E_{cs} = 28980 \text{ MPa} = 2,898 \cdot 10^7 \text{ KPa}$$

$$y_{cg} = \frac{S_x}{A} = \frac{\sum y_i \cdot A_i}{A} = \frac{2,5 \cdot (49-9) \cdot 5 + 6,5 \cdot 9 \cdot 13 + 11,5 \cdot 3 \cdot 3}{(49-9) \cdot 5 + 9 \cdot 13 + 3 \cdot 3} = \frac{1364}{326} = 4,18 \text{ cm}$$

$$I_{x'} = \frac{(49-9) \cdot 5^3}{3} + \frac{9 \cdot 13^3}{3} + \frac{3 \cdot 3^3}{12} + 3 \cdot 3 \cdot 11,5^2 = 9455 \text{ cm}^4$$

$$I_{x, cg} = 9455 - 326 \cdot 4,18^2 = 3759 \text{ cm}^4$$

$$3759 = \frac{49 \cdot h_e^3}{12} \quad h_e = \sqrt[3]{\frac{12 \cdot 3759}{49}} = 9,73 \text{ cm}$$

*altura equivalente
de massa de mesma rigidez (peso 2,43 kN/m²)*

$$D_e = \frac{E_{cs} h^3}{12(1-\nu^2)} = \frac{2,898 \cdot 10^7 \cdot 0,0973^3}{12(1-0,2^2)} = 2317 \text{ KNm}$$

para 7: $l_x = 3,85 \text{ m}$



$$f_{d,m} = g_k + \psi_2 g_k = 2,57 + 0,3 \cdot 1,5 = 3,02 \text{ kN/m}^2$$

$$w_0 = \frac{5}{384} \frac{f_{d,m} l_x^4}{D} = \frac{5}{384} \cdot \frac{3,02 \cdot 3,85^4}{2317} = 3,73 \cdot 10^{-3} \text{ m} = 3,73 \text{ mm}$$

$$w_{00} = (1 + \phi) w_0 = (1 + 2,5) \cdot 3,73 = 13,05 \text{ mm}$$

$$w_{adm} = \frac{l}{250} = \frac{3850}{250} = 15,4 \text{ mm}$$

Como $w_{00} = 13,05 \text{ mm} \leq w_{adm} = 15,4 \text{ mm}$,
a laje possui rigidez adequada

Estado Limite Último

Dimensionamento à flexão

Armadura mínima

$$A_{s, \min} = \rho_{\min} A_c = \frac{0,15}{100} \cdot 326 = 0,489 \text{ cm}^2/\text{viga}$$

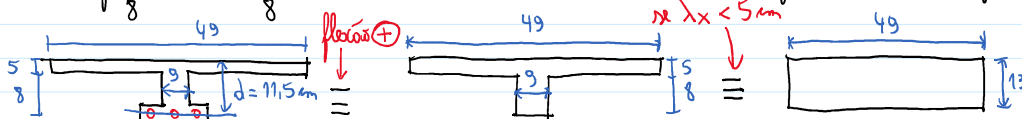
Momento fletor

$$\frac{\text{KNm}}{\text{m}} \cdot \frac{\text{m}}{\text{viga}}$$

$$f_d = 1,4 \cdot 2,57 + 1,4 \cdot 1,5 = 5,70 \text{ kN/m}^2 \quad \times 0,49 \text{ m/viga}$$

$$M_d = \frac{f_d l_x^2}{8} = \frac{5,70 \cdot 3,85^2}{8} = 10,56 \text{ KNm/m}$$

$$5,7 \text{ kN/m/viga} = 517 \text{ KNcm/viga}$$



Supondo vigas retangulares com largura $b = 49 \text{ cm}$

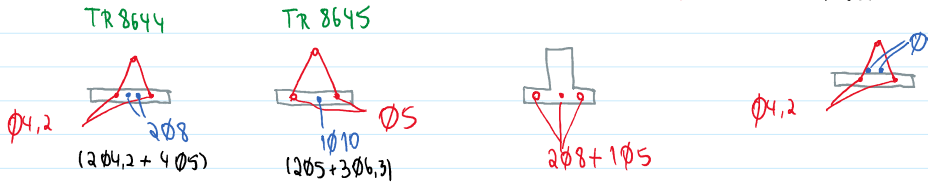
Supondo seção retangular com largura $b = 49 \text{ cm}$

$$\chi = \frac{d}{\lambda} \left(1 - \sqrt{1 - \frac{2M_d}{\alpha_c \eta_c f_{cd} b d^2}} \right) = \frac{11,5}{0,8} \left(1 - \sqrt{1 - \frac{2,517}{0,85 \cdot 10 \cdot (2,5/1,4) \cdot 49 \cdot 11,5^2}} \right) = 0,78 \text{ cm}$$

$$\chi x = 0,8 \cdot 0,78 = 0,62 \text{ cm} < h_f = 5 \text{ cm} \rightarrow \text{hipótese correta}$$

$$A_s = \frac{\alpha_c \eta_c f_{cd} b \chi x}{f_{yd}} = \frac{0,85 \cdot 10 \cdot (2,5/1,4) \cdot 49 \cdot 0,8 \cdot 0,78}{43,48} = 1,07 \text{ cm}^2/\text{vig} \checkmark$$

$$2\phi 4,2 + 2\phi 8: A_s = 2 \cdot \pi \cdot \frac{0,42^2}{4} + 2 \cdot \pi \cdot \frac{0,8^2}{4} = 1,28 \text{ cm}^2$$



Esforço constante

$$\text{Sobrecarga: } R_d = \frac{f_d \cdot l_x}{2} = \frac{5,70 \cdot 3,85}{2} = 10,97 \text{ KN/m} \quad \times 0,49 \text{ m/vig} \rightarrow 5,38 \text{ KN/vig}$$

Resistência:

$$\sigma_{Rd} = 0,25 f_{cd} = 0,25 \cdot 0,15 f_{ck}^{2/3} = 0,25 \cdot 0,15 \cdot 25^{2/3} = 0,32 \text{ MPa} = 0,032 \text{ KN/cm}^2$$

$$K = 1,6 - d = 1,6 - 0,115 = 1,485$$

$$2\phi 8 + 1\phi 5: A_s = 1,20 \text{ cm}^2 \quad \rho_1 = \frac{1,20}{9 \cdot 11,5} = 0,0116$$

$$V_{Rd1} = \sigma_{Ra} K (1,2 + 40 \rho_1) b_w d = 0,032 \cdot 1,485 (1,2 + 40 \cdot 0,0116) \cdot 9 \cdot 11,5 = 8,18 \text{ KN/vig}$$

$V_{sd} = 5,38 \text{ KN/vig} < V_{Rd} = 8,18 \text{ KN/vig}$, a larg. possui segurança adequada para o esforço constante.

Amplitude de distribuição

$$\left. \begin{array}{l} \text{NBR 14860-1} \\ \text{EF-96} \end{array} \right\} \begin{array}{l} A_{s, \text{dist}} = 0,60 \text{ cm}^2/\text{m} \\ A_{s, \text{dist}} = 5 \cdot \frac{h_f}{f_{yd}} = \frac{5 \cdot 5}{43,48} = 0,57 \text{ cm}^2/\text{m} \end{array} \left. \begin{array}{l} \phi 4,2 \leq 23 \text{ cm} \\ \phi 5 \leq 32 \text{ cm} \end{array} \right\} A_{s, \text{dist}} = 0,60 \text{ cm}^2/\text{m}$$

